DESIGN & ENGINEERING PORTFOLIO

INFERNO RACING

DESIGN.
DEVELOP.
DOMINATE.

Ansys · ROYAL ENFIELD · Lenovo · SHRI Agrawal Sweets & Namkeen SINCE 1956 · AiREN INTERNATIONAL WORLD-CLASS PACKAGING · THE SHISHUKUNJ INTERNATIONAL SCHOOL INDORE · zevar by SETHIS · tekzee technologies · CHOPADA MOTORS · BALAJI GROUP PROTECTING PEOPLE AND PLACES · raah foundation

RESEARCH OVERVIEW

TABLE OF CONTENTS:

2. RESEARCH OVERVIEW
3. RESEARCH OVERVIEW
4. DESIGN INSPIRATION
5. CAR BODY DEVELOPMENT
6. NOSE CONE DEVELOPMENT
7. REARWING ANALYSIS
8. WHEEL DEVELOPMENT
9. 3D MODEL
10. CNC TOOLPARTS
11. MANUFACTURING AND TESTING

GENERAL RESEARCH TOPICS

- Force calculations for FEA analysis
- Moment of inertia which axis
- Tipping moment and centre of mass
- Bluff body drag and wheel wake
- Surface smoothening and G continuity

Drag Force:

It refers to the resistance offered by the surrounding medium, which opposes the motion of the car. In research we found that one of the main factors that hugely affect the speed and working of the car is aerodynamic drag. From the formula $F = \frac{1}{2}\rho v^2 C_a A$, it can be derived that reducing Drag Coefficient and and Frontal Area of the car would be most relevant in terms reducing the drag of the car. Next step was to find out how we could reduce the drag coefficient and the frontal area of the car. For the frontal area aspect, it was key to reduce the surface area of the cone. For the drag coefficient aspect, it was key to make an aerodynamic car body

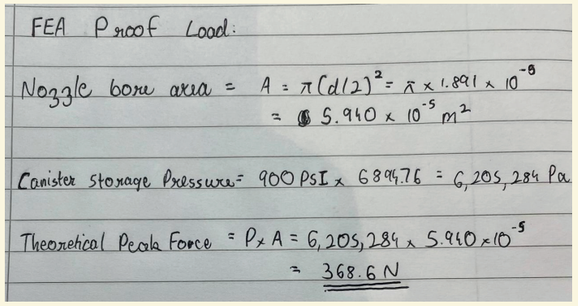
Aerodynamics:

After studying about drag force, we came across multiple aerodynamic factors that affect the drag coefficient and moreover the speed of the car

Bluff Body Drag- The main area for the bluff body drag was the halo, as wind passed through the steep angles of the halo, it was separated earlier than the trailing edge of the halo. This separation then caused a low pressure zone (wake) that created a suction force towards the low pressure area and created aerodynamic lift.

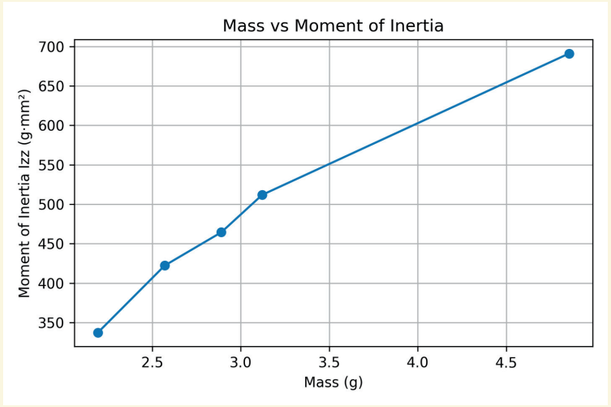
Moment of inertia: (formula image)

While researching we found that the moment of inertia of the wheel would affect the rotation of our wheel, with the help of the formula we understood that we could either reduce the radius of wheel/rim or the mass of the wheel. But due to a regulation cap for the diameter of the wheel, there wasn't much we could do about it, so our main focus was reducing mass. However while making multiple iterations of the wheel we also found out that there were certain areas of the wheels that had a greater effect on the moment of inertia (more specifically rim of the wheel) and the mass of the wheel. As seen from the graph, the moment of inertia is directly dependent on mass (values from our front wheels)



Surface Smoothing and Continuity:

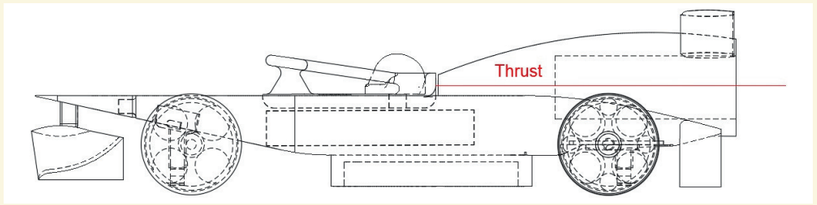
Controlling air flow is vital to reduce drag. However rough surfaces and irregular joints create turbulent air flow leading to wake regions. This was most commonly seen near the Halo. These surfaces or joints are classified as G0 continuity which meet at an angle and create a visible kink. On the other hand G1 continuity means that when two surfaces meet, their positions and tangent match at the connection point.



RESEARCH OVERVIEW

TIPPING MOMENT AND CENTRE OF MASS:

Tipping moment refers to the moment created by the distance between the thrust vector of a car and its centre of mass which tries to rotate the car on its wheels. This in particular has a graver consequence for f1 in school cars as they have a less mass so it is more sensitive to pitching moment. It also leads to a loss of energy indirectly as a portion of the propulsion energy is diverted to the rotational motion of the car body, this also leads to unbalanced load transfer between the wheels and further increasing the rolling rotation. Hence the key to reducing tipping moment was to reduce the distance between centre of gravity and the thrust vector.



ROLLING RESISTANCE AND BEARING FRICTION:

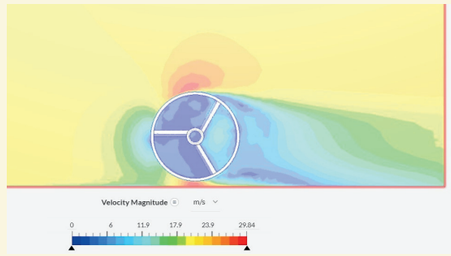
Aerodynamic drag is not the only force opposing the car. Rolling resistance acts at the contact patch and is caused by deformation of the wheel material, as material compressed at the front of the patch recovers at the rear, and because this deformation is not perfectly elastic a small amount of energy is lost as heat each rotation. The force is given by $F = C(r) \times N$, where $C(r)$ is the coefficient of rolling resistance and N is the normal force. Since normal force is set by the weight of the car, our focus was the contact patch itself: a harder, smoother surface deforms less and loses less energy, which is the reasoning behind regulation T7.7 requiring a consistent rolling surface with no tread, and why we sanded our wheels to a uniform finish after printing. The second loss is bearing friction, which directly opposes rotation and is compounded by the moment of inertia of the wheel, as a wheel resisting angular acceleration tends to skid rather than roll. Our regionals car suffered exactly this problem and incurred penalties, leading us to design a dedicated bearing seat into the hub of Wheel V3. Alignment matters equally, as any toe error scrubs the contact patch sideways and produces a resistive force far larger than rolling resistance alone, so we verified all four wheels were parallel and rotating freely using a 3 degree inclined plane test.

FORCE CALCULATION:

To estimate the amount of force each component of a car would withstand, we had to calculate the peak force. We measured the diameter of the nozzle to be 8.6974mm. However the peak force here was taken into account if full bore areas were open at full storage pressure simultaneously. However this condition is not practically possible due to the piercing mechanism of the gas canister and pressure decay. Hence, we had to reference from Hydron's Portfolio which came out to be an estimate 10N

WHEEL WAKE:

After observing cfd simulations, we noticed another area where low pressure zones (wake) were prominent. These areas were behind the wheel. After research we found out that this wake was created due to the rotation of the wheel which disturbed the airflow and created vortices and unsteady flow.



MATERIAL STRENGTH:

(Table) For manufacturing, we had 5 materials to choose from. To ensure we select the best material we had to compare the yield strength, young's modulus and the density of the 5 materials. The reason for these 3 criterias was to check the deformation and weight of the material.

Material	Yield Strength (MPa)	Young's Modulus (GPa)	Density (g/cm³)
ABS	20	2.24	1.06
PLA	46	1.85	1.015
PETG	40	2.1	1.27
Nylon	45	2.7	1.14
Carbon Fiber (Epoxy Composite)	300	133	1.43

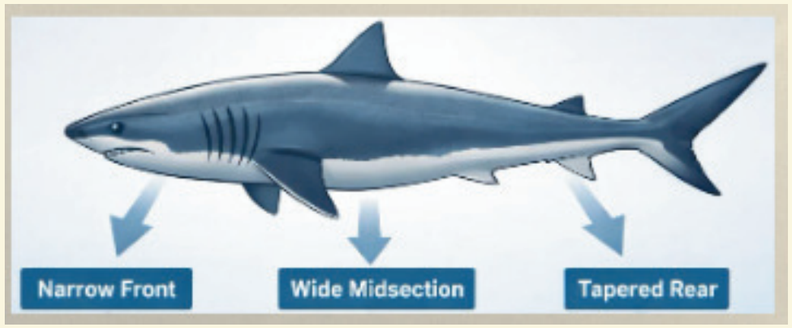
IMPULSE, MOMENTUM AND THE LAUNCH PHASE:

While drag determines how much the car slows during the run, the launch phase determines how fast it is travelling in the first place. From Newton's second law, $a = F/m$, acceleration is inversely proportional to mass, and since thrust is fixed by the standard 8g power pack, mass was the only variable we could control. Expressed as impulse, $Ft = m\Delta v$, so $\Delta v = Ft/m$ — for a fixed impulse, exit velocity is inversely proportional to mass. Any mass reduction therefore produces a direct gain in speed, which is why mass became a governing objective across every component, from cutting material out of the wheel rims to shelling the side pods. The cartridge also does not deliver constant force: thrust peaks immediately after piercing and decays as internal pressure falls, so most of the impulse is delivered within the first fraction of a second. This is why our 10N peak force estimate was used as the FEA design load, as it represents the worst case the structure experiences, and why the car spends most of the track decelerating under drag and rolling resistance rather than accelerating.

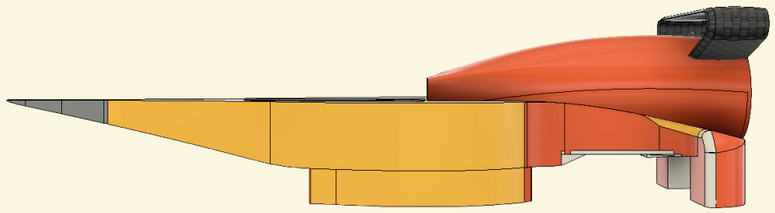
DESIGN INSPIRATION

Biomimetic roots- Shark Design

The design of the car body was inspired by the shape of a shark because a shark's body is very aerodynamic because it is able to cut through/travel through water very easily. The car, like a shark, has a narrow front like a hammerhead shark followed by a wide midsection and a tapered rear shape.



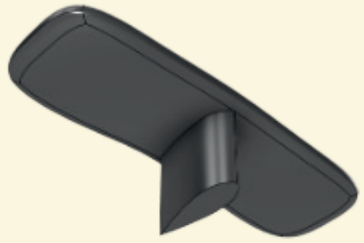
This is the car before the front wing was made as separate parts without the halo and the gas canister. We can see that the shapes are very similar, this is because the shark's body is designed to separate water cleanly at the front and the water stays attached along the body and finally leaves cleanly at the back. This property is also ideal for our car which flows through air minimizing drag.



Rear Wing Designs

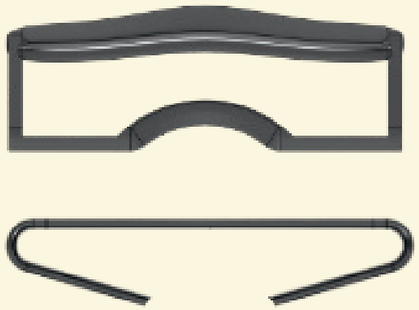
Water/ Tear Drop Shape

The support structure of the rear wing for the v2 car was inspired by a rain droplet. A water droplet has an almost spherical shape or a teardrop shape which is pointed from one end and rounded from the other. Due to surface tension, it shrinks to minimize surface area and takes the shape of a sphere. As we know drag is affected by frontal area so having minimum surface area is very important and hence the shape of the v2 rear wing support structure was like a rain droplet (teardrop shape) where the round surface was at the front so that it could separate wind smoothly and then the pointed/tapered end minimizes turbulent wake.



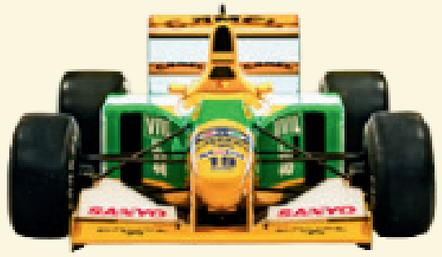
Triangular Shape of Rear Wing

Following the FEA results, the first rear wing had a rectangular shape which increased the amount of material used, hence we picked the shape of a triangle for the rear wing because it has only 3 edges so the amount of material used and surface area would decrease but it wouldn't affect the structural integrity as it is able to distribute force if applied on the wing to two different different directions or vertices of triangle



Inspiration from Benetton B192 1992 F1 Car:

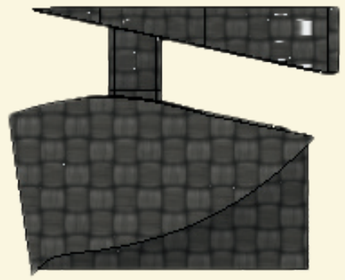
The nose cone of the car was inspired from the Benetton 1992 F1 car which was a revolutionary high nose designed front which created massive amounts of downforce. The front wing of the car has a flat, straight mainplane, this simplicity provided a stable aerodynamic platform, the mainplane was flanked with rectangular box endplates. Another feature was that the middle section was actually raised. Looking at its characteristics, it was perfect for us as the end plates separated the high pressure winds on the top from the low pressure winds beneath the nose cone and thus prevented lift-inducing vortices. Unlike a step design we created a ramp attached to the mainplate to direct the air above the wheels and prevent turbulent wakes being created near the wheels.



Kingfisher Beak – Nose Cone Profile

The kingfisher dives from air into water at high speed with almost no splash, because its long, narrow beak widens gradually so the water it displaces is pushed smoothly outward along the taper rather than compressed ahead of it as a pressure wave. Our car faces the same problem, as a blunt nose cone builds high pressure directly ahead of it, increasing drag and forcing air outward toward the front wheels. Following this principle, our nose cone tapers from a narrow tip and widens gradually rearward, separating air cleanly at the tip and guiding it along the body rather than pushing it ahead of the car.





CAR BODY DEVELOPMENT

Objectives:

- Achieve G1 continuity
- Develop a car body that has smooth air flow below ad and above the car while also reducing aerodynamic drag
- Ensure the entire car body complies with all STEM Racing Technical Regulations
- Maintain sufficient stiffness throughout the body, particularly around the rear side-pods and cartridge chamber

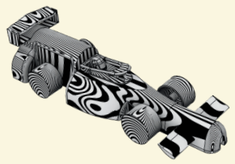
Surface Smoothness Evaluation:

Aim: Get regular and consistent patterns signifying G1 continuity (image of 1st car)

Zebra Analysis on V1:

Through this analysis we can find rough and obscure surfaces such as:

1. large surface blobs at the side pods which suggest significant curvature discontinuities
2. Nose cone area has completely chaotic zebra patterns around the front
3. Complete change in pattern from tapered front to main body where halo sits and then change in pattern from halo to gas canister, this means that there is a complete shift from one surface to another with minimal continuity

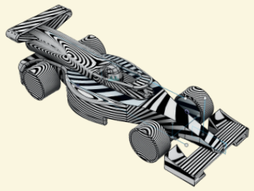


Evaluation:

Key takeaway from the Zebra Analysis was that there were two core areas where surfaces needed to be smoothen or connected better that were between the nose and main body, halo and the gas canister. G1 continuity was set as the standard for all aerodynamic joints for Final Car

Zebra Analysis on Regionals Car:

1. This pattern is quite different from V1 and changes we see are:
2. The Main body from the top surface has even stripes and a regular pattern which means it has G1 continuity
3. The transition between nose and main body is smoother than V1
4. Side pods have clean and parallel stripes flowing rearwards which is good.

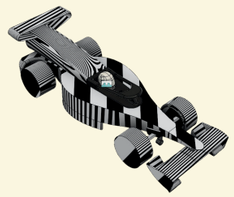


Evaluation:

There is an improvement over V1 but still some surfaces are filleted so the zebra patterns are irregular, and the transition between the nose and main car body is still discontinuous as evident by the change in stripes between their connection point

Zebra Analysis on Nationals Car: (image of nationals car)

1. Our National car had the most uniform zebra patterns out of all:
2. Nose and Car Body have a seamless connection with zebra stripes flowing through the car body into the nose cone without irregularities
3. Zebra Stripes are parallel and regular on the surface where the halo sits



Regulatory Compliance: Cartridge Chamber Safety Zone

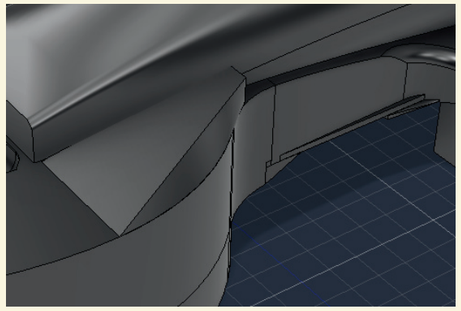
This led to us realising that the car had one major technical violation- the safety zone around the cartridge chamber. We used the loft tool to make our cartridge chamber in order to maintain an aero shape but after making it we only checked the safety area at the start and the end of the chamber not realising that the loft concaves in the middle leading to it having a thickness less than 3.5mm so we fixed this by changing the diameters of the original sketch objects used to create the loft.

Manufacturability & Structural Optimization: Sidepods & Rear Pods

The second problem was that we had made the side pods and the earpods for our car and we had made them hollow using the shell tool but we had made them too small to be properly cut by the cnc machine so to fix that we made the sidepods and the rear pods bigger but after shelling them when we ran the fea analysis we realised that the rear pods had a safety factor which was almost below target so we decided to make the rear pods whole and not shell them in order to ensure structural stability.

Aerodynamic Design Improvements

After we had aligned the car body with the nose cone for optimum aerodynamics and fixed all the violations and manufacturability issues with our car body we thought about other ways that the design could be improved even further



We also thought about adding small ramps to the side of the gas canister chamber to pust wind up from the rear wheels but that resulted in the car body being pushed down without having much effect on the rear wheels so we dropped that idea

High speed analysis showed that F1 in schools cars tend to lift when launching. This can be explained by looking at the thrust vector of the CO_2 cartridge. Inserted in the air cartridge chamber, the height and angle determines the vector of force. By changing the angle of the chamber

Other than parallel to the track surface, a small amount of energy is lost as shown in the graphic below. Therefore, we stuck to an angled chamber

Conclusion & Objective Evaluation

The final car body satisfied all four design objectives by integrating seamlessly with the redesigned nose cone, complying fully with the STEM Racing technical regulations, remaining manufacturable using CNC machining, and achieving the required structural integrity. The iterative optimisation process demonstrated that aerodynamic performance could only be maximised once manufacturing constraints, regulatory compliance and structural behaviour had all been considered simultaneously.

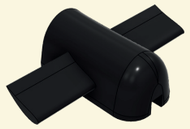
NOSE CONE DEVELOPMENT

Project Objectives

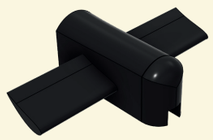
- Structural Integrity: Durable and sturdy so as to not break or flutter during the race.
- Flow Management: The nose cone should protect the wheels from drag and create proper airflow for the rest of the car.
- Aerodynamic Efficiency: The nose cone itself should be aerodynamic and not result in excessive drag.

Previous Designs:

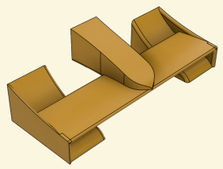
V1: Our first iteration of the car had a large frontal area because of the nose cone being very rounded and the front wings a bit behind the starting of the nose cone. This led to a high amount of drag force.



V2:- the modified nose cone or (v2) was tapered from the front to reduce its frontal area however we still weren't satisfied with the design as it was blunter and results weren't as positive as we were hoping.



Evaluation:- We realised that the problem wasn't only that nose cone's frontal area but also that the front wing placement was behind the starting of the nose cone.



Pre Nationals (V3):- So our third design of the nose cone had a very flat and tapered design along with having a front wing attached from the starting of the ne cone so as to successfully reduce the drag force.

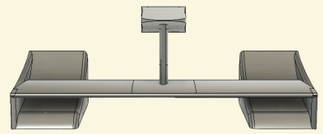
Analysis & Design Iteration

Before starting the design process we looked at the designs and results we had from our previous competition.
Instead of thinking of the nationals car as a separate project we decided to use the date we had from our previous competition which enabled us to build and improve from our previous design.
Firstly looking at the nose cone from our nationals qualifier car.
We realised that while we were protecting the wheels from drag using the nose cone, the cone itself was hitting a lot of air and pushing it up which was resulting in the car being pushed downwards.

Rethinking our approach of pushing the airflow upwards we decided the design our nose cone such that most of the air is tunneled through the bottom of the car instead of pushing it up, so instead of the earlier upward slope of the nose cone this time we decided to keep the top flat so as to interfere least with upwards airflow and give the nosecone's bottom a downward slope starting from the nosecone's front to its rear.

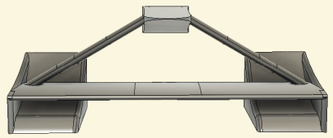
Design 1:

The first way that we joined the two bodies was through a singular perpendicular joint which was designed by extruding an elliptical sketch which is more aerodynamic than a cuboidal joint and more structurally safe than a cylindrical joint. After designing it we felt that the joint would interfere with airflow and decided to think about an alternate approach.



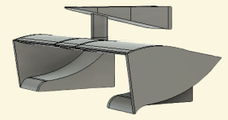
Design 2:

The second design used joints of the same shape which ensured structural stability and aero performance but instead of a single perpendicular joint this approach had two diagonal joints from the side of the noce cone to the upper surface of the front wing. We made this design keeping in mind our initial objective of increasing uninterrupted wind flow by tunnelling it through the bottom of the car as this design did not interfere with our central windflow tunnel by using diagonal joints which occupied the sides instead of the front.



Design 3:

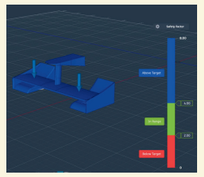
We created a new design which had one vertical joint and in this design we increased the height of the front wing so the length of the joint was decreased and there was clear airflow above and below the wings, we also changed the shape of the endplates to minimize weight.



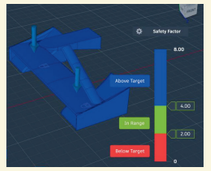


Finite Element Analysis (FEA) Evaluation

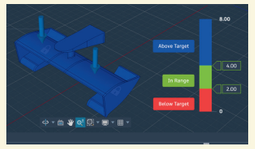
When finalising the design for our nose cone we first decided to run fea analysis on all three concepts to check if they were structurally sound as this was one of our main objectives.



Design 1: Had a minimum safety factor of 12.145



Design 2: Had a safety factor of 11.585



Design 3: had a safety factor of 9.9

Computational Fluid Dynamics (CFD) & Final Selection aerodynamic analysis:

We ran computational fluid analysis on the designs:
Design 1: the first design had a drag force of 0.17N
Design 2: the second design had a drag force of 0.17N
Design 3: the third design had a drag force of 0.15N
After looking at all 3 designs and the tests we reached the following conclusion-
Design 3 had the lowest drag force out of the three designs and was also compliant with all of the nosecone regulations while the other two designs that we had developed violated regulation T8.7, so we finalised design 3 for our nationals car.

REAR-WING ANALYSIS

Computational Fluid Dynamics (CFD) & Final Selection

Aerodynamic Analysis:

We ran computational fluid analysis on the designs:

Design1: the first design had a drag force of 0.17N

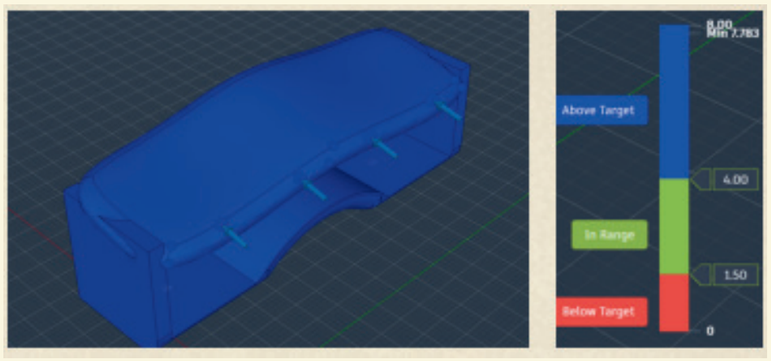
Design2: the second design had a drag force of 0.17N

Design3: the third design had a drag force of 0.15N

After looking at all 3 designs and the tests we reached the following conclusion-
Design 3 had the lowest drag force out of the three designs and was also compliant with all of the nosecone regulations while the other two designs that we had developed violated regulation T8.7, so we finalised design 3 for our nationals car.

AIM: Achieve Target Safety Factor of 1.5

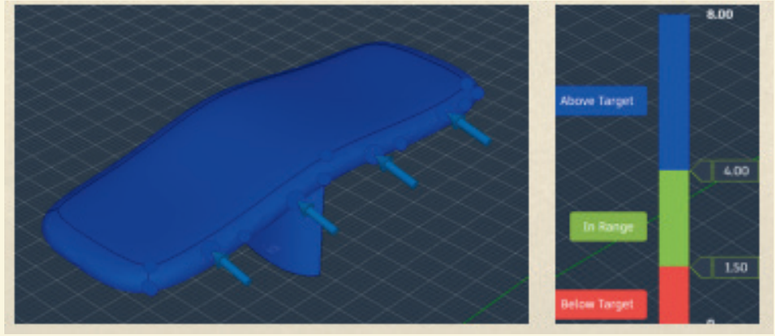
The results of rear wing of v1 car:
The results show a completely blue rear wing meaning that the structure is structurally sound with a Minimum Safety Factor of 7.783, But such a large difference between the TSF and MSF means that material is overused meaning weight of car and surface area would increase meaning more drag.



Evaluation:

We could reduce the thickness of the walls and supporting structure of the rear wing and it inspired the design for the rear wing of v2.

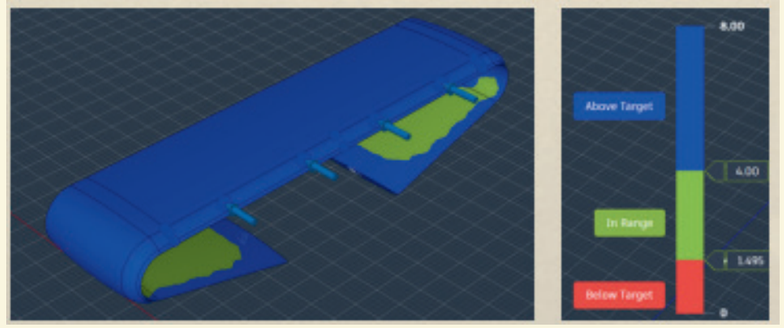
The results of rear wing of v2 car:
The rear wing had only a central supporting structure for the rear wing trying to minimize the material used. However the results were surprising as the Minimum Safety Factor was 18.2. This gave two conclusions, first that the structure of the v1 rear wing was actually very strong because of the amount of material used, not because it distributed force well. Second was that once again volume of material could be reduced.



Evaluation:

The design of the final rear wing of the V3 car was made keeping in mind conclusions of the FEA Analysis of v1 and v2 rear wing.

The results of rear wing of v3 car:
The rear wing was now thinned along with its support system to distribute force evenly or consistently. The results were exactly as what we were predicting as the Minimum Safety Factor was 1.48 achieving our aim of making a rear wing that structurally strong with minimum use of material so as to reduce drag and weight of the car



After looking at all 3 designs and the tests we reached the following conclusion-
Design 3 had the lowest drag force out of the three designs and was also compliant with all of the nosecone regulations while the other two designs that we had developed violated regulation T8.7, so we finalised design 3 for our nationals car.

WHEEL DEVELOPMENT

Parameters for tracking development of wheel-

- Moment of inertia- for the optimum rotational velocity of the wheel, we had to reduce the moment of inertia as much as possible
- FEA Analysis- We used Fusion 360's FEA analysis to measure the maximum deformation and stress the wheel could handle under 5 force
- Manufacturability and Compliance- After creations of designs, we checked whether the wheels were compliant with the regulations and manufacturable

Objectives of Development-

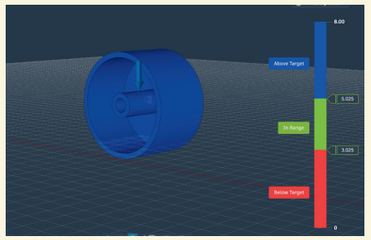
- Minimal Mass- through multiple iterations of cutting wheel mass by reducing radius and thickness of wheel, we found out that there was a direct relation between wheel mass and moment of inertia. Hence with every design, we tried to manipulate the radius and thickness of the wheel such that it was under regulations and also was rigid enough to not break under 50N force applied
- Yield Strength and Deformation- We tested our wheel designs by applying a 50N force, which was a proof load to understand stress concentrations and expose weak geometries. We targeted a factor of safety between 3-5. For deformation we targeted a peak deformation less than 0.05 mm which was 0.3% of the wheel radius to ensure deformation remained well below manufacturing tolerances and would not noticeably affect rolling accuracy.

Design Evaluation Process:

Wheel V1 (first iteration):

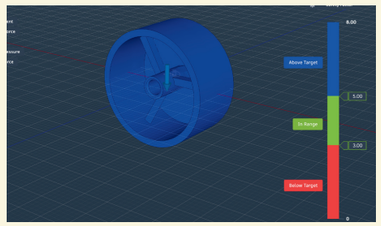
Our first wheel design was a basic design that included a simple hub attached to the
Peripheral of the wheel. This design was made to understand a regulation complaint wheel without keeping in mind the moment of inertia and mass of the wheel.

Results-
Moment of Inertia: 691.1
MSF: 32.44



Wheel V2 (Regionals Wheel):

We used this wheel for our regional competition, with its design having a 3 spoke structure for support.
We went with this design as it helped decrease the mass of the wheel and the moment of inertia by 100
units.



Results:
Moment of Inertia: 422.2
MSF: 10.79

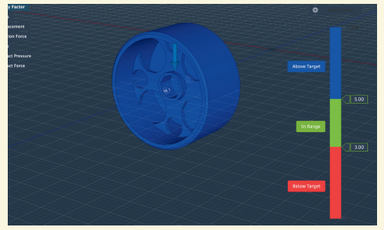
Defect in Regionals Wheel: Originally the diameter for the wheel was set to be 28.3 mm however with 3D printing inconsistencies, smoothening the wheel caused its diameter to reduce to less than the 28mm diameter required as per regulations

Results after adjustment:
Moment of Inertia: 511.9
MSF: 11.77

Wheel V3 (hole-design):

After regionals, we found out a major flaw in our wheel design which was not incorporating bearing into our wheels because our wheel did not rotate freely and we incurred penalties due it. Hence in this design we first made a bearing seat in the hub, then we changed the structure, having a 8 spoke web structure, to minimize weight while still having a strong support structure.

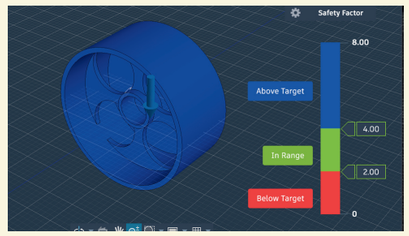
Results:
Moment of Inertia: 464.4
MSF: 12.09



Evaluation: After analysing and observing each wheel's fea analysis and results, we understood that the material used could be reduced and the moment of inertia was extremely high. So our next objective was to create a wheel that used optimal material volume so as to reduce moment of inertia. Also when we manufactured wheel V3, its support structure was so thin that it wasn't able to 3D print properly.

Wheel V4:

This wheel design was optimized to have minimum weight and moment of inertia, so we reduced the thickness of the wheel. We also redesigned the spoke structure, making 4 big holes in the support structure to improve structural integrity



Results:
Moment of Inertia: 337.3
MSF: 13.53

Selection of Wheel: After creating 5 iterations, we decided to choose wheel V4, as it had the least mass and moment of inertia as compared to the other, while following all the regulations and being manufacturable, fulfilling all our objectives.

3D MODELING

SOFTWAREUSED

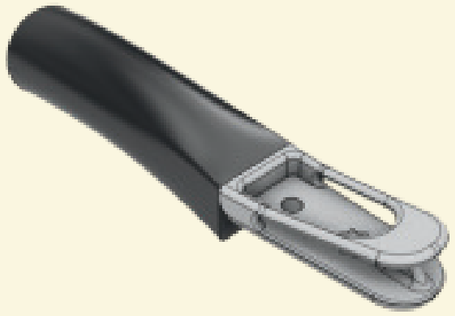
Fusion 360 was opted for making the car because, first it offered cloud based management where our engineers could easily work together on the platform and could view stored data.

Second was that it integrated CAD (Computer Aided Design), CAM (Computer Aided Manufacturing) and CAA (Computer Aided Analysis) eliminating our requirement for using multiple softwares to develop our car.
Also it was very accessible as we could use it on an educational license for free.

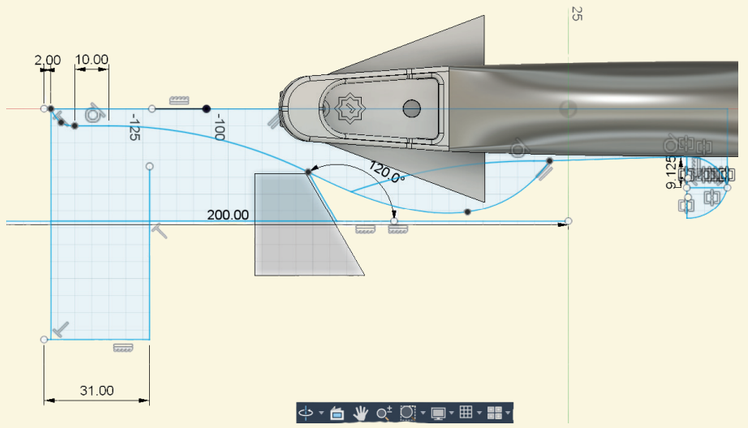
We used multiple tools to design our car, we started with basic sketch and extrude and fillet, once we had familiarized ourselves with these basics we started building the first version of our actual car.

We used the basics extrude, cut, fillet, along with the form tool to shape our body and rear wing. We also used the cut function to make space for the nose cone and the halo+helmet to be fit.

RnD and Analysis revealed thatthe car needed to be tapered.We used the spline tooltocutthenose cone intoa taperedshape andalsoused it to create a teardrop shape for the rear wing support structure .We also used create form to design our rear wing where we created 4 faces,andinclinedthewingby5degrees



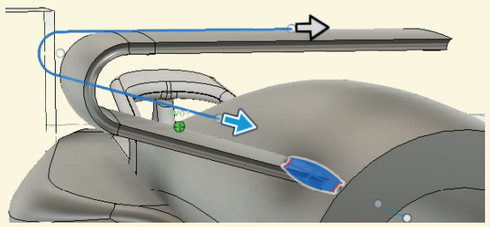
We used a fit-to-spline for the sketches of our car body to design the sidepods as they had to be curved.



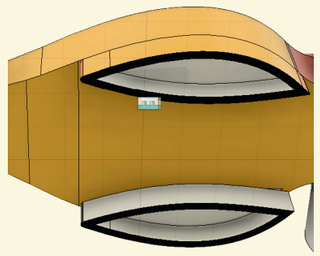
While these tools helped us build a car that was in compliancewiththe technicalregulations, the design seemed too simple and heavy to be our final design.

So we switched things up and built a new design from scratch; we not only used the basic tools that we had used for our initial concept but we also used other advanced tools that helped us refine our car while also keeping it light weight.
We used fillet in our sketches and the loft tool using the back face of the halo and two circles- one at the distance that would be the start of the gas canister and the other at the end of the gas canister with the circle sketch at the end of the halo being bigger to give the rear of the car an aerodynamic shape, this also created G1 surface continuity between the halo and rear car body.

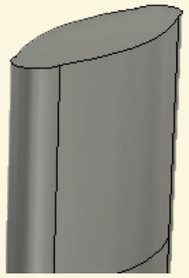
We used the sweep tool for making the rear wing of our car by making an elliptical sketch of the rear wing's aerodynamic shape and sketching out a triangular path for the sweep tool.



Further research conducted revealed that CoM needed to be shifted near the thrust vector, so to do that we had to reduce the weight of the car by using the shell tool.We shelled the side pods present at the underside of the car.



For ensuring that our joints throughout the car remained structurally safe as well aerodynamic we came up with a joint design which included an elliptical sketch being extruded.



MASS CALCULATION
It was important to get an idea of the mass of the car, to see whether the car was abiding by the regulations and was not underweight.To get a precise answer, we assigned materials to the car body, the rear wing, the nose cone, wheels, etc.
Rear Wheel support structure-2.61g
Front Wheel support structure-2.39g
Front wheels(2)-4.2g
Rear wheels(2)-5g
Rearwing-3.64g
Nose cone-6.34g
halo & helmet-9.5g
car body-22g

CNC TOOLPATHS

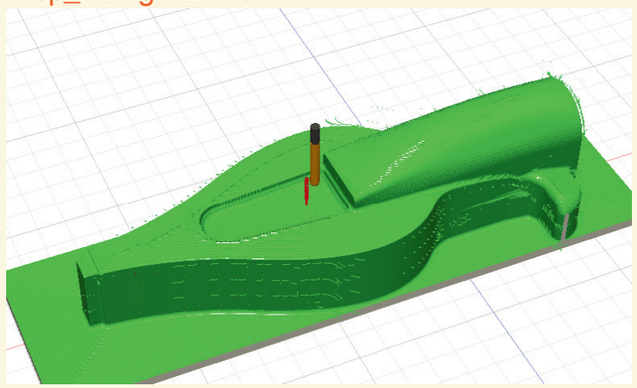
Cam and cnc-
For the cam and cnc processes we used the manufacturing workspace in Fusion 360 which is the same we used for designing the car and creating animations for marketing purposes.

Objectives-
Create proper toolpaths for the main carbody which is to be manufactured through CNC milling to ensure manufacturability
Create from the top, bottom and side of the car to ensure that the entire car body is manufacturable through CNC milling

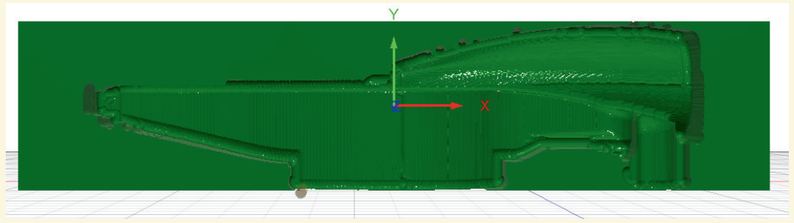
For the national qualifiers we had created the toolpaths from the top of the car, with the z axis of the mill facing downwards and that had given us confirmation that the top and the sides of the car were manufacturable but we did not know about the manufacturability of the bottom of the car that is where we faced problems when we got our manufactured car especially the side pods and the rear side pods so before we sent the car for manufacturing we also created the toolpaths from the bottom of the car which confirmed for us that the adjustments that we had to the side [ods by making them bigger and to the rear side pods by not shelling them and keeping them whole, our current design was manufacturable through CNC milling.

We created three separate setups by the name of- top_milling,bottom_milling, and side_milling and set the dimensions of the stock in each setup exactly according to the dimensions of the STEM Racing India Model block.
The three separate setups took an individual time as follows:

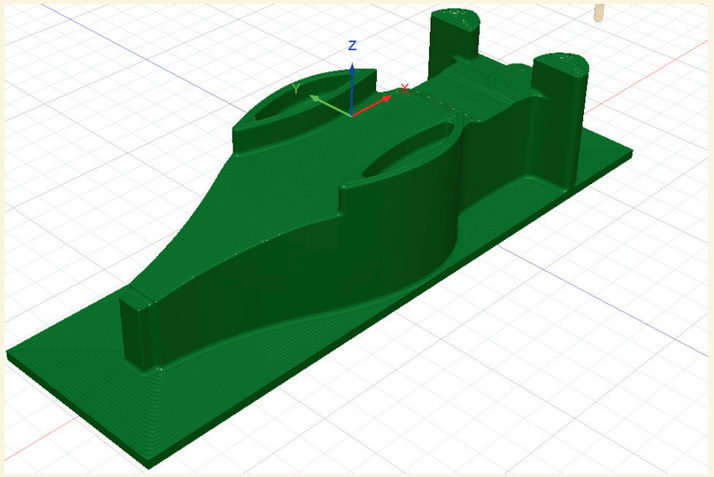
Top_milling-1:27:48



Side-_milling-00:25:36 for one side



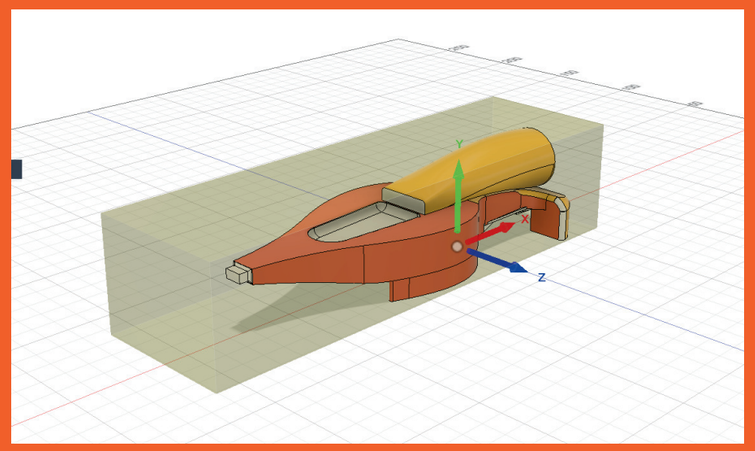
Bottom_milling-1:16:24



This brought our total toolpath time to about- 3:35:24 along with some additional times to complete other aspects of the car body like the gas canister chamber as what we have calculated is through our three simulations and represents an approximate time.

Using these three also helped us solve the manufacturing problems that we had faced with the national qualifier car as the bottom part of the car had not been manufactured correctly specifically the side pods that were not successfully milled as they were too small so sheeling them created cavities that the machine was unable to manufacture

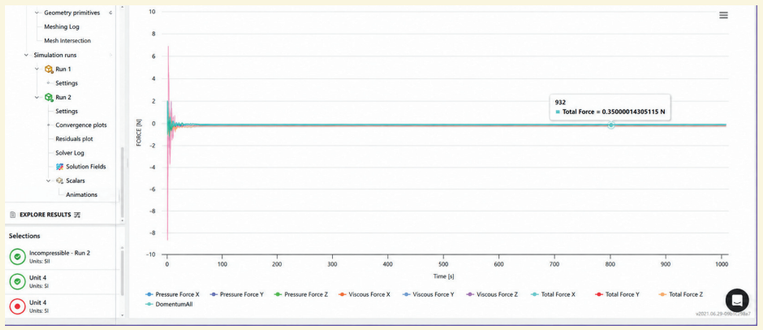
Process-
The IR-11 body was manufactured using CNC milling, It's machining workflow and simulation was planned in Fusion 360's Manufacturing itself. All dimensions had been computed in mm and a 3mm ball nose end mill was generated as the primary cutting tool to make the complex aerodynamic curves and contoured surfaces of the body.
We created three separate setups by the name of- top_milling,bottom_milling, and side_milling and set the dimensions of the stock in each setup exactly according to the dimensions of the STEM Racing India Model block.



MANUFACTURING & TESTING

Computer aided analysis and testing-

After we designed the entire car and had run fea analysis on each singular component and had created the toolpaths for the main car body to ensure manufacturability we ran computational fluid analysis on the entire car to find out if our car was performing well aerodynamically and how much better it was as compared to our national qualifier car.
We set the velocity of the computational fluid analysis at 22 m/s to get an accurate value of the drag force that the car would generate during racetime.



The drag force of the total car came out to be 0.35N which was a considerable improvement from our first version of the car result of 0.69N

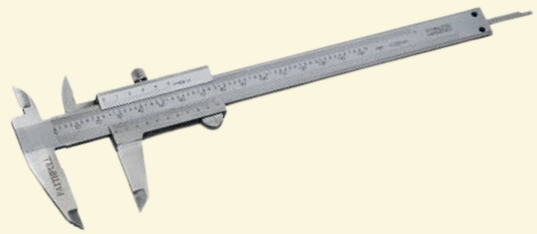
Manufacturing:

For the manufacturing of the model block we gave the order to STEMplify as we realized that the level of precision that we required from the car would not be delivered by local vendors as STEMplify has machines that are specifically intended for the purpose of CNC milling and STEM racing components.
For 3d printing the car components we gave the order to a 3d printing company based in our city, Indore and used the machine

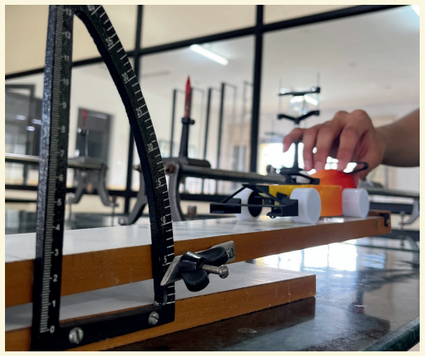
The main car body that we received was almost identical to our design and the results that we had gotten from the manufacturing toolpaths in fusion 360 so the car body needed very little sanding to smooth over the grooves caused by milling. machine

Physical Testing:

For the testing of the car we used instruments available in our school physics lab:
We used a vernier calliper with a least count of 0.01 cm. We used the vernier calliper to ensure that all parts of the car were within the minimum and maximum measurements as specified in the technical regulations rulebook.



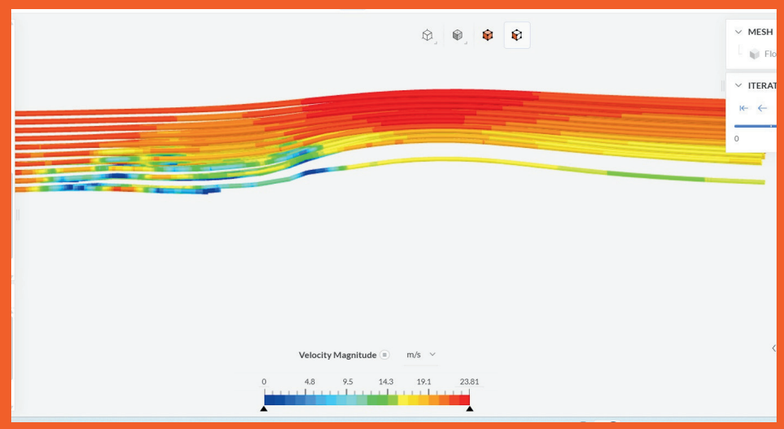
For ensuring that our car was rolling on a 3 degree incline plane we used an inclined plane with precise angle measurement to ensure this.



To ensure the safety of the tether line guides we suspended a hooked weight of 200g on the tether line guide.
When our car passed all of these tests we were able to determine that our car was race ready and safe to race.

Streamline Analysis – Airflow Over the Final Car

We ran a velocity streamline simulation on the assembled car to confirm our surface work was producing the flow behaviour we designed for, as zebra analysis proves our surfaces are tangent continuous but not that the air follows them. The upper streamlines remain smooth, parallel and attached at 21–24 m/s with no divergence or curling, showing the air is following the surface rather than separating and that pressure drag is minimised. This validates the G1 continuity targeted at the nose-to-body and halo-to-body joins, as a discontinuous surface would have separated at those transitions. The lower streamlines slow near the front of the underbody as expected, but recover downstream rather than stagnating, confirming the air is channelled through the underbody tunnel instead of forming a trapped low pressure region.



Manufacturing Safety-

We took multiple safety measures to ensure that the assembly and finishing process of our car was completely safe:

Safety Equipment	Function
Safety Gloves	Protect hands from sharp edges of the CNC-machined part and cutting tools during handling
Safety Goggles	Shield eyes from dust and spray paint during the sanding and painting the car
Dust Mask / Respirator	Prevent inhalation of fine particles and dust generated while sanding the body and spray painting the car
Workshop Apron	Shield clothing and body from dust,spray paint, super glue splatter during assembly